Post-Quantum Secure Communication for Autonomous Vehicle Platoons

Batch/Group/Team No.: 11

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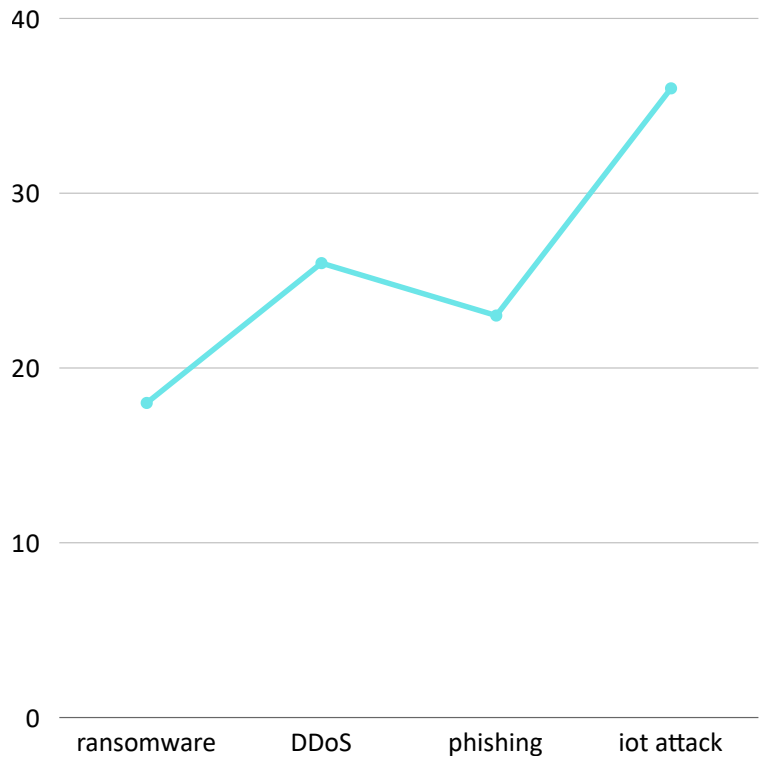
Motivation

- Over 9 in 10 vehicle networks rely on RSA/ECC, posing risks in a quantum future.
- Platoons exchange 10–50 critical messages per second, demanding ultra-secure communication.
- Transportation systems face hundreds of cyber incidents each year, exposing security gaps.
- This drives the need for a quantum-resilient communication framework for safer, future-ready mobility.



SDG Alignment & IKS Integration

- **SDG Alignment:**
  - **SDG 9 – Industry, Innovation, and Infrastructure:** Secure and resilient intelligent transportation systems.
  - **SDG 11 – Sustainable Cities and Communities:** Improved road safety and reliable autonomous mobility.
- **Indian Knowledge System (IKS):**
  - **Raksha (Protection)** : Proactive safeguarding of safety-critical transportation systems through quantum-resilient security.
  - **Anvikshiki (Scientific Inquiry)** : Application of analytical reasoning and cryptographic innovation to solve emerging cybersecurity challenges.

Literature Survey (Existing Technology)

Ref.No	Title, Author, Year	Methodology Used	Dataset / Evaluation	Advantages	Disadvantages
[1]	Quantum-resistant Transport Layer Security, García et al., 2024	Hybrid TLS integrating Kyber and Dilithium with modified OpenSSL prototype	Measured handshake latency, overhead, and CPU usage in a controlled network setup	Strong quantum resistance; practical protocol implementation	Higher overhead; not designed for vehicular environments
[2]	On Hardware Acceleration of Quantum-Resistant FOTA, Perazzo, 2024	Experimental implementation on automotive hardware	Firmware signing benchmarks	Real-world OTA feasibility; implementation-focused	Emphasizes signatures more than V2V communication
[3]	Efficient Implementation of CRYSTALS-Kyber on ESP32, 2025	Embedded system implementation	Runtime and memory analysis	Shows PQ feasibility on constrained devices	Hardware-specific results
[4]	Open Quantum Safe: Practical Quantum-Safe Cryptography, Stebila et al., 2020	Library-based implementation and TLS integration	Handshake performance tests	Production-ready crypto framework	Requires integration effort

Literature Survey (Existing Technology)

Ref.No	Title, Author, Year	Methodology Used	Dataset / Evaluation	Advantages	Disadvantages
[5]	CRYSTALS-Kyber: A CCA-Secure Module-Lattice-Based KEM, Bos et al., 2018	Algorithm design with reference implementation	Security and performance benchmarks	NIST-selected; strong security	More theoretical than system-level
[6]	CRYSTALS-Dilithium: Digital Signatures from Module Lattices, Ducas et al., 2018	Signature scheme with implementation notes	Computational analysis	Quantum-resistant authentication	Larger signature sizes
[7]	Post-Quantum Security for IoT Devices, Alkim et al., 2016	Implementation study on embedded platforms	Device-level benchmarks	Demonstrates real-world deployment	Older performance data
[8]	Secure Vehicular Communication Using Cryptographic Techniques: A Survey, Lenka & Rath (India), 2019	Threat and protocol analysis	Literature-based	Highlights vehicular security needs	Does not address PQ cryptography
[9]	3GPP LTE vs IEEE 802.11p for Vehicular Safety, Vinel, 2012	Network simulation	Vehicular traffic scenarios	Identifies low-latency requirements	Focus on classical security

Objective Of The Project

- To develop and evaluate a post-quantum cryptographic architecture that enhances confidentiality, authentication, and system resilience in autonomous vehicle platoon networks.

Objectives & Scope

- Secure V2V communication against quantum attacks
- Enable low-latency encrypted platoon messaging
- Protect OTA updates with quantum-safe authentication
- Build a scalable, future-ready security framework
- **In Scope**
  - Lattice-based key exchange
  - Quantum-resistant digital signatures
  - Hybrid cryptographic model
- **Out Scope**
  - City-scale scalability
  - Real-world road testing
  - Legal / regulatory certification

Proposed System / Methodology

1. OTA Server

- Sends signed firmware updates using Dilithium via liboqs.
- Vehicles install updates only after verification.

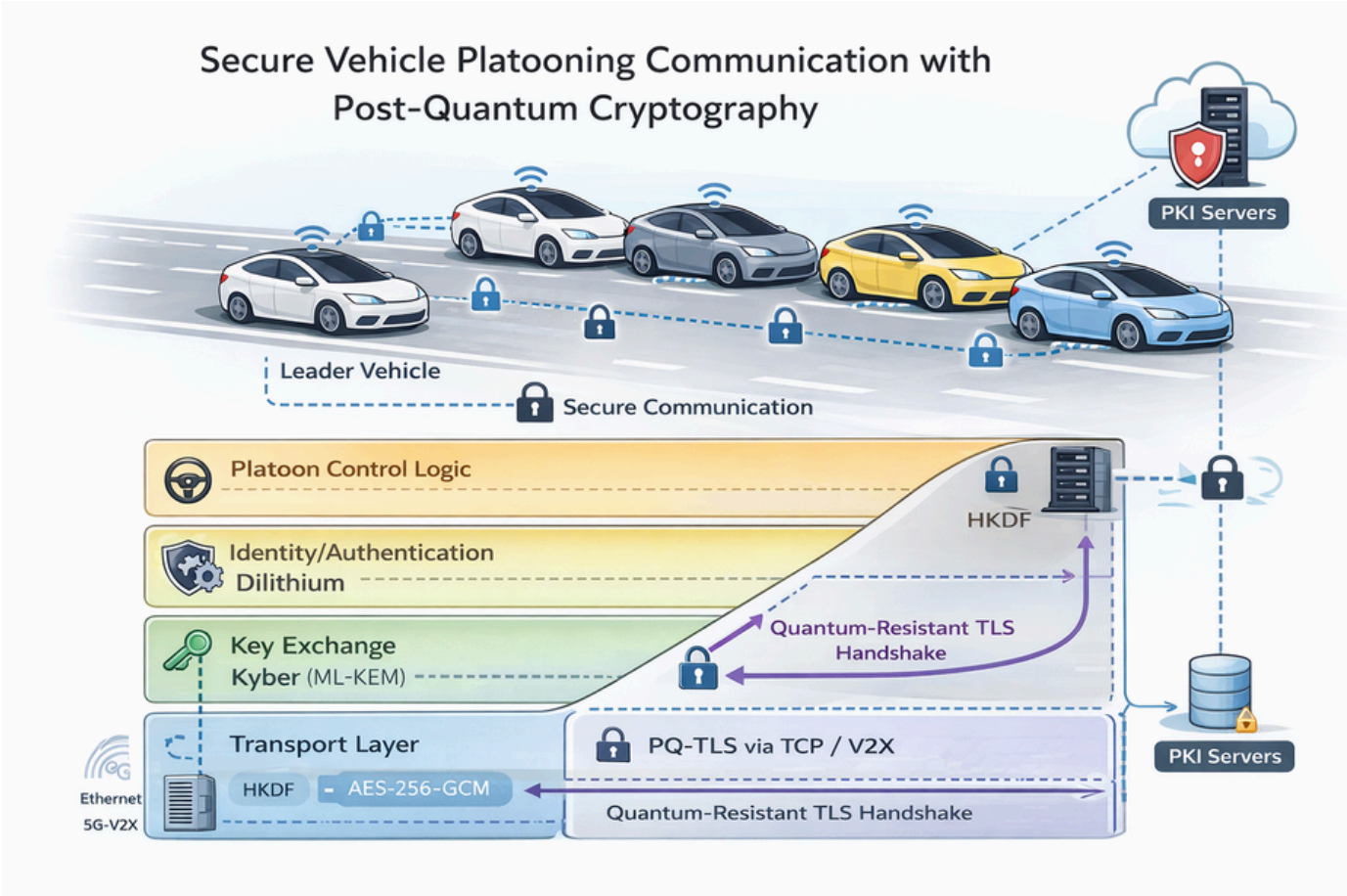
2. Vehicle Node

- Uses post-quantum identity keys and Kyber for secure key exchange.
- HKDF derives a session key.
- TLS encrypts messages using cryptography.

3. Secure V2V Communication

- Vehicles authenticate, establish a secure session, exchange encrypted data, preventing impersonation and tampering.

Proposed System / Methodology



Comparative Analysis & Novelty

Novelty:

- Presents a pure post-quantum architecture designed for autonomous vehicles, embedding quantum-resistant security into both communication and software updates.

Feasibility:

- Achievable with mature libraries like liboqs and cryptography, and compatible with modern vehicular hardware.

Project Plan

Phase	Core Goal	Key Technologies
Phase 1 (Sem 6)	Identity & Secret Sharing	Kyber, HKDF, TLS
Phase 2 (Sem 7)	Trust & Integrity	Dilithium, liboqs, OTA Server
Phase 3 (Sem 8)	Agility & Validation	Hybrid Protocols, Benchmarking

Expected Outcome & Resources

Final Deliverables:

Winning a Hackathon:

- Develop an innovative solution under time constraints, demonstrating creativity and technical skill. Recognition validates the idea's real-world impact.

Publishing a Paper:

- Share research and findings in an academic format to contribute to the field. Enhances credibility and supports future advancements.

Patenting the Idea:

- Secure intellectual property rights to protect the invention. Enables commercialization and partnership opportunities.

Expected Outcome & Resources

Tools / Libraries (Open-Source & AICTE Recommended):

- Python — Core development and protocol implementation.
- liboqs — Open-source library for post-quantum cryptography (Kyber, Dilithium).
- cryptography — Secure encryption primitives including AES-GCM.
- Git — Version control system for collaborative development.

References & Review Queries

- C. R. García, J. A. Jiménez, M. E. Rojas, and F. H. Calderón, “Quantum-Resistant Transport Layer Security (TLS),” Journal of Systems Architecture, vol. 144, p. 102585, 2023, Elsevier, doi: 10.1016/j.sysarc.2023.102585.
- National Institute of Standards and Technology, “Post-Quantum Cryptography Standardization,” NIST, Gaithersburg, MD, USA, 2024.
- J. Bos et al., “CRYSTALS-Kyber: A CCA-secure module-lattice-based KEM,” in Proc. IEEE European Symposium on Security and Privacy (EuroS&P), 2018, pp. 353–367.
- L. Ducas et al., “CRYSTALS-Dilithium: A lattice-based digital signature scheme,” IACR Trans. Cryptographic Hardware and Embedded Systems, vol. 2018, no. 1, pp. 238–268, 2018.

References & Review Queries

1. Why was Kyber selected for key exchange over other post-quantum algorithms?
2. How does the system handle latency and computational overhead in real-time vehicular communication?
3. What mechanisms prevent attacks such as impersonation, replay, and message tampering?
4. Is the architecture scalable and adaptable to future cryptographic standards and vehicular hardware?

Thank you